Pre-review-book DLBCSM201

Lesson 1 - Introduction to Matrices

1.1 Basic concepts of matrices

Question 1

Level of difficulty: E

Display the following table as a graph.

	North	Middle	South
Product A	120	150	100
Product B	277	259	310

Question 2

Level of difficulty: E

For the following matrix A = $\begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}$, specify the transposed matrix.

Solution question 1

Solution hint: In a matrix, there are no header rows and no margin columns.

Short solution: $\begin{pmatrix} 120 & 150 & 100 \\ 277 & 259 & 310 \end{pmatrix}$

Detailed solution: The numbers are transferred in exactly the same order as in the rows and columns, excluding the header row and margin column: $\begin{pmatrix} 120 & 150 & 100 \\ 277 & 259 & 310 \end{pmatrix}$

Solution question 2

Solution hint: Columns and rows are swapped.

Short solution: $A^T = \begin{pmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{pmatrix}$

Detailed solution: The columns and rows are swapped, $A^T = \begin{pmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{pmatrix}$

Question 3

Level of difficulty: M

Specify one column vector and one row vector each.

Question 4

Level of difficulty: M

Determine the main diagonal for the given matrix. Indicate the type of the matrix. Does the matrix represent a special form? If so, what is its name?

A = $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 8 \\ 0 & 0 & 9 \\ 0 & 0 & 0 \end{pmatrix}$

Solution question 3

Solution hint: Row vectors are to be represented as $\vec{v}^T$.

Short solution: Column vector $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, row vector $\vec{b}^T = (4\ 5\ 6)$

Detailed solution: With the column vector, only one column is represented; with the row vector, only one row is represented. Column vector $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, row vector $\vec{b}^T = (4\ 5\ 6)$.

Solution question 4

Solution hint: Consider the diagonal elements and the values that are above and below the diagonals. The type is mxn.

Short solution: $a_{11} = 1$, $a_{22} = 7$, $a_{33} = 9$, type 4x3, upper triangular matrix

Detailed solution: The diagonal elements are $a_{11} = 1$, $a_{22} = 7$, $a_{33} = 9$. The matrix has four rows and three columns, so the type is 4x3. Since the values are above the diagonals, it is called an upper triangular matrix.

Question 5

Level of difficulty: D

Consider whether the following statements are false:

- If A^T is a diagonal matrix, then A is also a diagonal matrix.
- If A is a zero matrix, then A^T is also a zero matrix.

Solution question 5

Solution hint: Take a close look at the structure of the matrix.

Short solution: Both statements are true.

Detailed solution: Transposing a diagonal matrix preserves the property of a diagonal matrix. If A is a zero matrix, then it contains only zeros. Thus, the transposed matrix is also a zero matrix.

1.2 Addition of matrices

Question 6

Level of difficulty: E

Given the following matrices A and B, calculate A + B.

A = $\begin{pmatrix} -1 & 4 & 6 \\ 2 & -3 & 7 \end{pmatrix}$ B = $\begin{pmatrix} 2 & 3 & 5 \\ 0 & 4 & 7 \end{pmatrix}$

Question 7

Level of difficulty: E

Given the following matrices A and B, is the arithmetic operation A – B possible? If yes, determine the result.

A = $\begin{pmatrix} 4 & 2 & 7 \\ 3 & 1 & 0 \end{pmatrix}$ B = $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$

Solution question 6

Solution hint: Add up the individual elements.

Short solution: C = $\begin{pmatrix} 1 & 7 & 11 \\ 2 & 1 & 14 \end{pmatrix}$

Detailed solution: Add up each of the elements of the matrices, e.g., c_{11} = -1 + 2 = 1.
C = $\begin{pmatrix} 1 & 7 & 11 \\ 2 & 1 & 14 \end{pmatrix}$

Solution question 7

Solution hint: Check the type of the matrices.

Short solution: Arithmetic operation not possible.

Detailed solution: Matrix A is a 2x3 matrix, matrix B is a 2x2 matrix. The matrices must be of the same type in order to add them.

Question 8

Level of difficulty: M

Given the matrices A = $\begin{pmatrix} 1 & 3 & 2 \\ -3 & 4 & 5 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & -1 \\ 0 & 4 \\ 1 & 5 \end{pmatrix}$, calculate A – B^T.

Question 9

Level of difficulty: M

Given the lower triangular matrices below, show that the sum of both is also a lower triangular matrix.

$A = \begin{pmatrix} -1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 4 & 1 \end{pmatrix}$, B = $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

Solution question 8

Solution hint: First determine B^T.

Short solution: A – $B^T = \begin{pmatrix} 1 & 3 & 1 \\ -2 & 0 & 0 \end{pmatrix}$

Detailed solution: B^T = - $\begin{pmatrix} 2 & 0 & 1 \\ -1 & 4 & 5 \end{pmatrix}$. A – B^T = $\begin{pmatrix} 1 & 3 & 2 \\ -3 & 4 & 5 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 1 \\ -1 & 4 & 5 \end{pmatrix}$ = $\begin{pmatrix} -1 & 3 & 1 \\ -2 & 0 & 0 \end{pmatrix}$

Solution question 9

Solution hint: Add both matrices.

Short solution: C = $\begin{pmatrix} 0 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 2 \end{pmatrix}$

Detailed solution: A + B = $\begin{pmatrix} 0 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 2 \end{pmatrix}$. Since only non-zero numbers are below the diagonal, the sum is also a lower triangular matrix.

Question 10

Level of difficulty: D

Given the two diagonal matrices A = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ and B = $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, show that A + B^T also results in a diagonal matrix.

Solution question 10

Solution: First, determine B^T. Then add A and B^T.

Short solution: C = A + B^T = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 9 \end{pmatrix}$

Detailed solution: B^T = $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$. C = A + B^T = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 9 \end{pmatrix}$. The new matrix is also a diagonal matrix.

1.3 Scalar multiplication and product

Question 11

Level of difficulty: E

Given the matrices A = $\begin{pmatrix} 0 & 2 & -1 \\ 1 & 4 & 5 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$, calculate 2 · A + B.

Question 12

Level of difficulty: E

Given the vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$, calculate the scalar product $\vec{a}^{\mathsf{T}} \cdot \vec{b}$.

Solution question 11

Solution hint: First, determine 2 · A.

Short solution: C = $\begin{pmatrix} 2 & 7 & 2 \\ 4 & 9 & 10 \end{pmatrix}$

Detailed solution: 2A + B = 2 $\cdot \begin{pmatrix} 0 & 2 & -1 \\ 1 & 4 & 5 \end{pmatrix}$ + $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$ = $\begin{pmatrix} 0 & 4 & -2 \\ 2 & 8 & 10 \end{pmatrix}$ + $\begin{pmatrix} 2 & 3 & 4 \\ 2 & 1 & 0 \end{pmatrix}$ = $\begin{pmatrix} 2 & 7 & 2 \\ 4 & 9 & 10 \end{pmatrix}$

Solution question 12

Solution hint: First, determine $\vec{a}^{\mathsf{T}}$. The result is a scalar.

Short solution: $\vec{a}^{\mathsf{T}} \cdot \vec{b}$ = 23

Detailed solution: $\vec{a}^{\mathsf{T}} \cdot \vec{b}$ = $(1\ 2\ 3) \cdot \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$ = 0 + 8 + 15 = 23

Question 13

Level of difficulty: M

Given the two matrices A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, calculate $2 \cdot A - 3 \cdot B^T$.

Question 14

Level of difficulty: M

Given the vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, and $\vec{c} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, check whether the following arithmetic operations are defined. If so, state the result.

a) $\vec{a} \cdot \vec{b}$ b) $\vec{b} \cdot \vec{c}$ c) $\vec{b}^T \cdot \vec{c}$

Solution question 13

Solution hint: First determine B^T. Then calculate $2 \cdot A$ and $3 \cdot B^T$.

Short solution: $2 \cdot A - 3 \cdot B^T = \begin{pmatrix} -6 & 2 & 4 \\ -7 & -5 & 3 \end{pmatrix}$

Detailed solution: $B^T = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 1 \end{pmatrix}$, $3 \cdot B^T = \begin{pmatrix} 6 & 0 & 0 \\ 3 & 9 & 3 \end{pmatrix}$, $\cdot A = \begin{pmatrix} 0 & 2 & 4 \\ -4 & 4 & 6 \end{pmatrix}$, $2 \cdot A - 3 \cdot B^T = \begin{pmatrix} 0 & 2 & 4 \\ -4 & 4 & 6 \end{pmatrix} - \begin{pmatrix} 6 & 0 & 0 \\ 3 & 9 & 3 \end{pmatrix} = \begin{pmatrix} -6 & 2 & 4 \\ -7 & -5 & 3 \end{pmatrix}$

Solution question 14

Solution hint: Consider the type and definition of the scalar product.

Short answer: a) no b) no c) yes: $\vec{b}^T \cdot \vec{c} = 14$

Detailed solution: a) Both vectors are not of the same type. b) Both vectors are of the same type, but two column vectors cannot be multiplied together. c) Yes: $\vec{b}^T \cdot \vec{c} = (3\ 2) \cdot \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 12 + 2 = 14$

Question 15

Level of difficulty: D

Given the matrix A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$, can the matrix be represented as the sum of a symmetric matrix and a skew-symmetric matrix?

Solution question 15

Solution hint: Consider the type of the matrix.

Short solution: No

Detailed solution: The matrix cannot be represented as the sum of a symmetric and skew-symmetric matrix because it is not square.